
Subject: Re: Vertex Identity Update
Date: Friday, June 12, 2026 at 11:54:23 AM Eastern Daylight Time
From: Anna Elizabeth Connelly
To: George Stermann
Attachments: chalkboard_notes.pdf

Sending you this PDF of your chalkboard notes in case you want it for your own records.

From: Anna Elizabeth Connelly <annaelizabeth.connelly@stonybrook.edu>
Date: Friday, June 12, 2026 at 10:52 AM
To: George Stermann <george.sterman@stonybrook.edu>
Subject: Re: Vertex Identity Update

3pm works for me!

On Fri, Jun 12, 2026 at 10:35AM George Stermann <george.sterman@stonybrook.edu> wrote:
Still on for 3 today? Or another time this afternoon?

On Wed, Jun 10, 2026 at 4:52PM Anna Elizabeth Connelly
<annaelizabeth.connelly@stonybrook.edu> wrote:

Dear George,

Thank you so much for the clarification! Very helpful.

Anna

From: George Stermann <george.sterman@stonybrook.edu>
Date: Wednesday, June 10, 2026 at 4:34 PM
To: Anna Elizabeth Connelly <annaelizabeth.connelly@stonybrook.edu>
Subject: Re: Vertex Identity Update

On Wed, Jun 10, 2026 at 3:17PM Anna Elizabeth Connelly
<annaelizabeth.connelly@stonybrook.edu> wrote:

Screenshot 2026-06-10 at 2.43.16 PM.png



HI, ANNA. SEE BELOW IN CAPS. HOPE THIS HELPS. PICTURES LOOK GREAT.
GEORGE

Dear George,

I am sorry in advance if the answers to these questions are really obvious.

That said, here we have a couple different notations going on and I am just trying to clarify/understand what I am seeing.

Type 1: dashed line + arrow THIS IS SUPPOSED TO MEAN CONTRACTING THE

VECTOR'S MOMENTUM INTO THE VERTEX. THE DASHED LINE REPRESENTS JUST A "SCALAR" $1/P^2$ TYPE OF PROPAGATOR.



Type 2: boson/wavy line + arrow I THINK THIS IS A FOUR-POINT FUNCTION BETWEEN VECTORS, WHERE ONE THE MOMENTUM OF ONE OF THE VECTORS IS CONTRACTED ONTO THE VERTEX. THIS COMES INTO THE WARD IDENTITY FOR THE Z-BOSONS.



Then, we also have two different propagator types going on:

Type 1: arrows pointing toward vertices on either side THIS IS SUPPOSED TO REPRESENT JUST THE "PROCA" PART OF THE UNITARY GAUGE PROPAGATOR, LIKE $K^\mu K^\nu / [M^2(K^2 - M^2)]$



Type 2: arrow pointing toward just one vertex WELL, IN THIS PICTURE, I THINK THE OTHER ARROW IS JUST MISSING!



I think that I might be getting a little confused about the difference in meaning between: physical /longitudinal/scalar polarization.

My thought was that longitudinal refers to when the particle has that the polarization in the direction of motion, and I thought that scalar polarization referred to when the polarization in that direction is proportional to the momentum of the particle. However, you then sometimes refer to “physical” polarization as well and I get confused again. I think I am just having a minor confusion. I am also not completely clear on when we decide that which particle/s in a given diagram/process has this high energy property.

FOR A MASSIVE VECTOR PARTICLE, THERE ARE THREE "PHYSICAL" POLARIZATIONS, AND ONE IS LONGITUDINAL. AT HIGH ENERGY, THE LONGITUDINAL ONE IS ALMOST EQUAL TO THE MOMENTUM, BUT ITS SCALAR PRODUCT WITH THE MOMENTUM IS STILL ZERO. THE "SCALAR" POLARIZATION EXACTLY EQUALS THE MOMENTUM/MASS, AND ITS SCALAR PRODUCT WITH ANY OF THE PHYSICAL POLARIZATIONS IS EXACTLY ZERO.

I was also thinking I should go through the calculation again for getting process W and Z bosons from the goldstone modes.(I am sorry for any incorrect physics I have written here so far.)

In productive news: I made a lot of progress on our diagrams and should be able to get the write up done by our Friday meeting once I clarify these confusions.

OK GREAT!

Here are some of examples of diagrams we can now use.

These are iterations of my cross-box, triangle-contact (I am not sure if there is a better name for this already) three-point vertex, and box diagram. The three-point vertex also has that “blank circle” option that allows us to be more general about which boson we are discussing. The current labelling is just to show capacity and is not meant to be physically correct. I also included a latex snippet, just for fun, so you

can see how nice and easy the macros are to work with.

Screenshot 2026-06-10 at 2.55.48 PM.png



Screenshot 2026-06-10 at 3.05.10 PM.png



Screenshot 2026-06-10 at 3.06.55 PM.png



As always, let me know if you have any questions, comments, or concerns.

Screenshot 2026-06-10 at 3.09.09 PM.png



Sincerely,
Anna